

Optimally scheduling and loading tow trains of in-plant milk-run delivery for mixed-model assembly lines

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Abstract

Purpose – This paper aims to investigate the scheduling and loading problems of tow trains for mixed-model assembly lines (MMALs). An in-plant milk-run delivery model has been formulated to minimize total line-side inventory for all stations over the planning horizon by specifying the departure time, parts quantity of each delivery and the destination station.

Design/methodology/approach – An immune clonal selection algorithm (ICSA) combined with neighborhood search (NS) and simulated annealing (SA) operators, which is called the NSICSA algorithm, is developed, possessing the global search ability of ICSA, the ability of SA for escaping local optimum and the deep search ability of NS to get better solutions.

Findings – The modifications have overcome the deficiency of insufficient local search and deepened the search depth of the original metaheuristic. Meanwhile, good approximate solutions are obtained in small-, medium- and large-scale instances. Furthermore, inventory peaks are in control according to computational results, proving the effectiveness of the mathematical model.

Research limitations/implications – This study works out only if there is no breakdown of tow trains. The current work contributes to the in-plant milk-run delivery scheduling for MMALs, and it can be modified to deal with similar part feeding problems.

Originality/value – The capacity limit of line-side inventory for workstations as well as no stock-outs rules are taken into account, and the scheduling and loading problems are solved satisfactorily for the part distribution of MMALs.

Keywords Scheduling, Mixed-model assembly lines, Line-side inventory, Loading, Material delivery, Tow trains

Paper type Research paper

Nomenclature

S	= integer parameter: set of stations (index $s \in S$);	B_s	= integer parameter: maximum capacity of inventory level at station s (in number of bins);
C	= integer parameter: number of work cycles in the planning horizon (index $c = 1, 2, \dots, C$);	d_{sc}	= integer parameter: number of bins required at station s in work cycle c ;
K	= integer variable: number of tours (index $k = 1, 2, \dots, K$);	A	= integer parameter: maximum capacity that can be loaded onto tow trains (in number of bins);
R	= continuous parameter: running time of tow trains for a round trip;	$D(c, s)$	= integer parameter: cumulative number of bins of parts in demand at station s in work cycle c ;
δ	= continuous parameter: replenishment time of tow trains in the supermarket between two tours;	$O(c, s)$	= integer parameter: cumulative number of bins of parts delivered to station s in work cycle c ;
p_s	= continuous parameter: the travelling time from the supermarket to station, including the loading and unloading time;	$IL(s, 0)$	= constant parameter: initial inventory level at station s ;
		$IL(s, c)$	= integer variable: inventory level at station s in work cycle c (in number of bins);

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b_{ks}	= integer variable: the number of bins a tow train takes to station s on tour k ;
t_k	= continuous parameter: the start time from the supermarket for a tow train on tour k ;
$(t_k, b_{k1}, b_{k2}, \dots, b_{k S })$	= a delivery task for a tow train; and
$\Omega = \{(t_k, b_{k1}, b_{k2}, \dots, b_{k S })\}$	= an integrated material handling scheme.

1. Introduction

Over the past few decades, the pressure of automobile industry has been growing as requirements for automobile products have been gradually increasing with the development of economy. To satisfy diversified demands of individual customers, mixed-model assembly lines (MMALs) have been widely adopted in today's automobile production (Shahabudeen *et al.*, 2002; Wang *et al.*, 2011). Under this circumstance, several models and different types of products are assembled on the same assembly line, and thus a variety of parts required in assembly line are indispensable. Because the available space is extremely scarce (Peng and Zhou, 2018), it is impossible to store all those parts at the factory in practical applications, let alone at assembly workstations. Therefore, enormous amount and diversity of parts have to be transported just in time (JIT) to multitudinous stations while avoiding shortages, otherwise any shortage of parts might lead to line stoppage and hundreds of assembly workers being idle (Zhou and Peng, 2017; Boysen *et al.*, 2015).

In such a context, the in-plant milk-run handling strategy has been used by an increasing number of automobile manufacturers for timely and accurate delivery of diversified parts to multiple workstations. In-plant milk-run systems are transportation systems, where materials are delivered from a central storage area to different destinations on predetermined routes in short intervals (Klenk *et al.*, 2015). This naturally puts pressure on the delivery as it requires a massive amount of frequent small-lot material handling. As a consequence, to maintain the MMALs well supplied and ensure the smooth operation of production, JIT supermarkets gradually take the place of centralized warehouses. Within this concept, the supermarket is a decentralized logistics area for stacking materials by the side of an assembly line, in which materials are replenished by industrial trucks in large lots from the centralized warehouse. When a workstation that the supermarket is in charge of requires materials, corresponding bins will be in preparation. These bins contain complete kits of parts for specific models. The parts are placed in the same order as the workpiece assembled along the line to reduce unproductive searching and walking time of assembly workers. (Bozer and McGinnis, 1992; Limère *et al.*, 2012). Moreover, these comparatively small and standardized bins can be stored in accessible racks (Finnsgård *et al.*, 2011), which leads to the ergonomic improvement.

Afterward, bins are transported to workstations by tow trains, which are wagons with electric towing equipment. First, the tow train is loaded in the supermarket with bins of materials required by workstations. Then it sets out to deliver the corresponding number of bins in sequence to workstations on the MMAL that are in demand of materials. Finally, at the end of each tour, the tow train returns to the supermarket to be replenished for the preparation of the next delivery.

When it comes to tow trains, it should be noted that they usually operate on a fixed schedule, and the tow train scheduling is within the scope of in-house logistics. As Emde *et al.* (2012) pointed out that the planning and control of this in-house logistics concept amounted to a complex task where the following interrelated decision problems had to be solved:

- 1 decision on the number and location of decentralized supermarkets;
- 2 assignment of line segments to supermarkets and determine the number of tow trains per supermarket;
- 3 determination of each tow train's fixed delivery schedule; and
- 4 decision on the bins to be loaded per tour of a tow train.

The first two problems are related to the supermarket, such as the supermarket location determination and deciding the number of tow trains per supermarket. The last two problems are related to the tow train, which means the frequency and time of tow trains leaving the supermarket to tour stations on its given route, the tow train stopping at which station and the number of bins it unloading there. As is known to all, a reasonable and comprehensive schedule plan of material delivery can definitely ensure a reliable and steady part supply, but it is hard to be achieved without taking into account the number of part bins to be loaded to corresponding stations per tour. Moreover, a well-scheduled plan can reduce the line-side inventory effectively, and because space on the assembly shop is scarce and expensive (Emde and Boysen, 2012), it is necessary to develop a scheduling plan for tow trains and determine the number of bins to be loaded on a tow train in each tour. In other words, the paper focuses on decisions 3 and 4. The tow train scheduling and loading problems are the most crucial ones in the field of part feeding. Similar to the digital factory that uses the simulation techniques to realize the design and optimization of production processes and system (Centobelli *et al.*, 2016; Kuhn, 2006), the scheduling and loading problems mentioned before use the simulation method to improve the scheduling process of the material delivery, so that the corresponding evaluation objective is optimized, which can make the optimization of the whole scheduling system come true. It is of great significance in the comprehensive process management, the model planning (Zakoldaev *et al.*, 2019), reducing costs of the manufacturing system and improving the core competitiveness of the enterprise. An efficient material handling system can effectively reduce the operating cost by 20–30 per cent (Zhou and Peng, 2019), second only to the optimal assembly sequence, which can reduce the total cost by 40–60 per cent (Bahubalendruni and Biswal, 2018; Bahubalendruni and Biswal, 2017). Therefore, it is fairly essential to do some research on the scheduling and loading problems.

To cope with the scheduling and loading problem of the material handling for MMALs, this paper adopts the immune clonal selection algorithm (ICSA) as the basis. The contributions of the paper are summarized as follows:

- 1 A material handling scheduling and loading method for MMALs based on the milk-run delivery with the constraints of capacity of the tow train and the line-side inventory. The objective is to minimize total line-side inventory for stations over the planning horizon while ensuring that materials are delivered in time so that the stock-out of stations can be eliminated.
- 2 Several lemmas and theorems of the research problem are proposed and proven to improve the quality of solutions and reduce computational complexity to some extent.
- 3 The intelligent ICSA incorporated with the SA and neighborhood search (NS) operators, which the authors call NSICSA, is used to tackle the problem, and its effectiveness and competitiveness are proved by comparing with other two algorithms.

The remainder of this paper is organized as follows. Section 2 provides a brief literature review. Section 3 describes the problem formulations of the scheduling problem based on in-plant milk run delivery. To solve the proposed problems, three theorems were put forward. Section 4 presents the NSICSA, which overcomes deficiencies of the SICSA, such as the tendency to trap into local optima and limited search depth. Computational experiments are carried out to evaluate and validate the proposed algorithm in Section 5. Finally, Section 6 draws conclusions.

2. Literature review

Although tow trains are widely applied in the automotive industry, there have been few relevant literatures tackling the scheduling of tow trains. As mentioned before, [Emde et al. \(2012\)](#) distinguished between four decision problems related to organizing the part feeding process with tow trains. To date, literatures involving the above four decision problems are introduced.

Regarding the first decision problem, namely, the supermarket location problem, [Battini et al. \(2012\)](#) described the supermarket concept explicitly and did surveys on basic decision problems emerging when implanting the supermarket concept. [Emde and Boysen \(2012\)](#) studied the supermarket location problem with the consideration of optimizing the supermarket configuration costs and total material flows, and a dynamic programming algorithm was obtained. However, the research did not take the capacity limit of the supermarket and unavailable areas in the workshop into consideration. Afterward, [Alnahhal and Noche \(2015\)](#) dealt with the two aspects, and the time complexity of the SLP problem was increased. Later, a real genetic algorithm (RGA) was carried out to handle the problem.

The second, third and fourth decisions, namely, the routing, scheduling and loading of tow trains, are the subproblems of the delivery-to-line (DL) problem. According to different handling vehicles, the DL problem can be divided into two categories: part distribution by the forklift (DL1) and material delivery by tow train (DL2). The two problem types correspond to the point-to-point and the milk-run delivery.

About the DL1 problem, [Kozan \(2000\)](#) studied the assignment (DL1-assignment) and sequencing (DL1-sequencing) problem of material delivery tasks for the truck assembly plant in Australia. The optimization model proposed by the authors is an extended application of the traditional multi-period vehicle routing problem (VRP). [Boysen and Bock \(2010\)](#) focused on Hoist's material delivery sequencing problem (DL1-sequencing), aiming at minimizing the maximum inventory in all stations and production cycles with part-dependent weights, and constructed a dynamic programming algorithm and SA algorithm. [Rao et al. \(2013\)](#) investigated the scheduling of a single vehicle. To avoid material shortage and reduce total costs including inventory holding cost and travelling cost, it was necessary to determine the departure time and part quantity of each delivery according to the predetermined assembly sequence. Despite the reference value of aforementioned literature, it is unrealistic for their models to be directly applied to the JIT material handling of MMALs.

The second, third and fourth decisions are relevant to the part feeding of assembly lines, which are consistent with material delivery by tow train (DL2). The modeling and optimization problems of material handling for MMALs in this paper have certain correlations and similarities with the traditional VRP ([Ho and Haugland, 2004](#); [Baker and Ayechev, 2003](#); [Laporte et al., 1988](#)) and inventory routing problem (IRP) ([Kleywegt et al., 2002](#); [Vidović et al., 2014](#)), but there are three differences:

- 1 Customers in the VRP are visited only once, unlike the part feeding in assembly lines that requires multiple delivery.
- 2 The classic VRP and its extension problems only deal with path optimization problems without considering the end demand.
- 3 The end demand of IRP-related problems is set to a certain value (or subject to a certain distribution), but the end demand of the material delivery of assembly lines problem is a time series.

With regard to the VRP problem, as for the milk-run delivery, [Vaidyanathan et al. \(1999\)](#) extended the classical VRP model and proposed a JIT capacitated vehicle routing problem (JITCVRP) to handle the routing problem of tow trains (DL2-routing). The authors assumed that once a route was assigned to a tow train, there was no interruption and the demand rate of each workstation was constant. To ensure the punctual delivery, the optimization objective was to minimize total driving time. The quantity of materials delivered each time before the next one reaching the workstation was sufficient to meet the demand of the station. Therefore, shorter delivery cycle meant less line-side inventory. Similar to the VRP problem, the DL2 problem was analyzed by [Emde and Boysen \(2010\)](#). They dealt with the routing of tow trains (DL2-routing) and the scheduling of material delivery (DL2-scheduling), and developed a nest dynamic programming algorithm to minimize the sum of the line-side inventory cost and travelling cost. On the basis, [Emde et al. \(2012\)](#) focused on the loading problem of tow trains (DL2-loading), determining the replenishment of each workstation with the predetermined maximum capacity of tow trains, demand of workstations and

cycle time of each tour. Thus, the bi-objective optimization of minimizing the maximum inventory and total inventory was accomplished with the construction of an exact polynomial-time approach. Golz *et al.* (2012) took a German automobile manufacturer as an example and proposed a heuristic solution procedure to investigate the routing problem of tow trains (DL2-routing), the scheduling of material delivery (DL2-scheduling) and the loading problem of tow trains (DL2-loading) simultaneously, aiming at minimizing the required number of shuttle drivers without stock-outs. Furthermore, Choi and Lee (2002) analyzed the joint optimization problem of the three as well and developed a two-stage heuristic algorithm to minimize the deviation of optimal handling time. According to the above discussion, these research studies have provided the authors a useful reference to resolve scheduling and loading problems of in-plant milk-run delivery for MMALs.

Because the scheduling and loading problems are proven to be NP-hard (Fathi *et al.*, 2015), exact approaches only fit small-scale problems. To settle large-scale problems with NP-hardness nature within desirable time, many researchers have resorted to metaheuristic algorithms which possess the outstanding computational efficiency and superior solution quality. For instance, a novel SA algorithm-based heuristic (Fathi *et al.*, 2014a), a novel memetic ant colony optimization-based heuristic algorithm (Fathi *et al.*, 2014b) and a modified particle swarm optimization algorithm (Fathi *et al.*, 2015) were proposed to tackle the scheduling and loading problems of tow trains, all aiming at minimizing the number of tours and line-side inventory. In addition, Peng and Zhou (2018) used artificial bee colony-based approach to investigate the JIT in-house logistics problem for automotive assembly lines. The multi-objective ant colony algorithm (Rubaiee and Yildirim, 2019) was developed to minimize total completion time and energy cost on a single-machine preemptive scheduling. Besides, facing the fourth industrial revolution, an improved binary grey wolf optimizer was proposed by Jiang *et al.* (2019) to solve the dependent task-scheduling problem.

Apart from the algorithms mentioned above, there are also a bunch of algorithms derived from or similar to the classic genetic algorithm (GA) which generates new candidates by crossover and mutation operators, such as differential evolution (DE) algorithm and ICSA etc. For example, Rao *et al.* (2013) developed a hybrid genetic algorithm and simulated annealing (GASA) approach to deal with the scheduling problem with high efficiency even in large instances. Moreover, an elitism genetic algorithm (EGA) combining an elitism strategy was proposed by Xie and Chen (2018) address uncertainty job shop scheduling problem with interval grey processing time. It can be seen that these algorithms show remarkable performance in search ability, short computational time and the extensibility, which provides inspirations that metaheuristic algorithms can be combined with other factors to obtain stronger effectiveness. Among these algorithms, the ICSA, which was designed based on the clonal selection of biological immune system, retains an outstanding information processing and problem-solving ability, which are embodied in the characteristics of distributed, self-learning, self-adaptation, self-organization and robustness. It is capable of converging to the global optimal solution rapidly in the searching process. Owing to its outstanding global search

capability, the ICSA has attracted great attention and been applied to tackle a number of optimization problems in scientific and engineering applications (Zhou and Wu, 2019), including robotic assembly line balancing and flow shop scheduling. For instance, regarding the flow shop decision-making problem, Liu *et al.* (2009) applied the ICSA to handle the hybrid flow-shop scheduling with the objective of minimizing makespan. Peng and Zhou (2018) solved the JIT part distribution problem for automobile assembly lines by the proposed modified ICSA. Zhou and Wu (2019) developed an improved immune clonal selection algorithm (MOICSA) to solve bi-objective robotic assemble line balancing problems, and the MOICSA showed great search and convergence ability, which led to better results compared to the other algorithms. As mentioned before, the purpose of the scheduling and loading problems tackled in this paper are meant to minimize total line-side inventory of all stations over the planning horizon, so it can be seen as a typical optimization problem. In addition, to the best of the authors' knowledge, ICSA has not been used to tackle the tow train scheduling and loading problems in the literature. Consequently, it is worthier evaluating the ICSA's capability of solving the problem tackled in this paper as a global search algorithm.

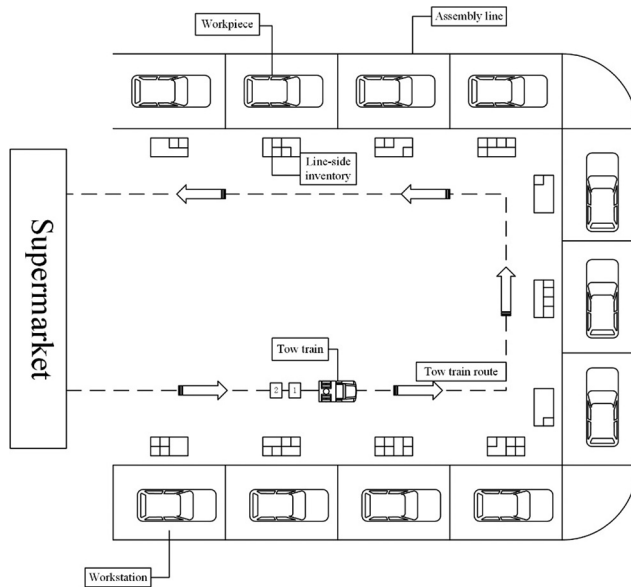
However, despite the abovementioned advantages, ICSA falls short in the local search phase, so in this paper, a novel intelligent ICSA incorporated with the SA and NS operators, which the authors call the NSICSA, is developed to solve the scheduling and loading problems of tow trains with line-side and tow train inventory capacity constraints. The proposed NSICSA possesses the global search ability of ICSA, the ability of getting rid of local optimum of SA and the deep search ability of NS. Owing to comprehensive advantages, it can enhance the quality of solutions, effectively solve the problems of optimization with constraints and has a fast convergence speed. Experimental results show that the NSICSA is of better performance compared with the ICSA in that it has the advantage of higher quality solutions and outstanding searching efficiency as well as avoiding prematurely converging to local optimum by means of chaos-based initialization.

3. Problem statement

3.1 Problem description and notations

Figure 1 reveals a representative layout of an assembly line with a supermarket described in the paper, and for the sake of convenience, the notations and explanations are generalized in notations and explanations. Using these notations, the scheduling and loading problem can be described as follows. In the in-plant milk-run delivery system for the automotive MMAL addressed in this paper, tow trains are used to make a frequent small-lot material handling to stations, which is denoted by $s \in S$ in the assembly line. Typically, a tow train carrying materials packaged in uniform standard-size bins sets off from a central supermarket to visit these stations in a fixed order (Emde and Gendreau, 2017). The production sequence is predetermined a few weeks before production, thus whether materials are required at station s in work cycle c ($d_{sc} = 1$) or not ($d_{sc} = 0$) is known with certainty. Because all bins are of the same capacity and size, the exact contents of bins are insignificant for the scheduling and loading problem. In this

Figure 1 Illustration of milk-run delivery system for automotive MMALs



case, the proper number of bins to be delivered to the accurate corresponding stations is a matter of concern.

The tow train with full bins travels from the supermarket to station s takes a certain amount of time p_s , including the loading and unloading time. It then travels around the assembly area, driving by all stations on the route and returning to the supermarket as an end, which takes time R . Moreover, the replenishment time of tow trains in the supermarket between two tours is δ . All times are normalized to the cycle time and all the time related parameters can be fractional. However, for convenience's sake, it is assumed that the tow train only sets off at integral points in time, namely, full work cycles (Emde and Gendreau, 2017), which has been proven to be a tolerable simplification. In addition, from a practical perspective, it is hard to communicate an accurate and continuous timetable to operators when the demand occurs owing to the higher precision.

Furthermore, given the limited space at stations, the maximum capacity of inventory level at station s is B_s , and tow trains are usually restricted to a small number of wagons and thus leads to constraints on the number of bins being loaded on a tow train.

In addition, two auxiliary functions are defined as follows for the convenience of the model formulation:

$$\mu_1(u, v) = \begin{cases} 1 & u \leq v \\ 0 & u > v \end{cases}, \quad \mu_2(u, v) = \begin{cases} 1 & u = v \\ 0 & u \neq v \end{cases}$$

In this paper, the objective of the scheduling problem is to minimize the total line-side inventory for all stations over the planning horizon. Clearly, multiple factors can affect the stock level at workstations. However, the departure time of each cycle and the number of bins for respective stations can be regarded

as the most important issues in the tow train scheduling to prevent the stock-out of workstations.

3.2 Mathematical model

In this subsection, a mathematical model is established on the basis of the following assumptions:

- Cycle time is selected as the basic unit time for the system.
- No tow train idle time is allowed between two consecutive jobs. Note that if idle time is allowed, more degrees of freedom concerning the objective function will exist. To avoid extra investment costs, a no-wait policy is applied.
- The location of the material supermarket and the workstation is predetermined. In addition, the routes and driving distances of each tow train are known.
- The processing time of part deliveries by tow trains from the supermarket to each station as well as the operating time for picking up and dropping off bins are static and deterministic.
- The production sequence is determined a few weeks before production, thus the demand of each station within planning horizon is known with certainty.
- Materials are stored in bins of uniform size, thus, the capacity constraint of tow trains and the inventory level at workstations are denoted by the number of standardized bins.

According to the assumption (equation 2), because of the no-wait policy as well as the material delivery for automotive assembly lines on the basis of in-plant milk-run delivery, the cycle of each handling task is $(R + \delta)$, that is to say, the total number of deliveries K in the planning horizon is fixed. Concerning the objective of material handling, the classic objective of minimizing total cost including inventory cost and transportation cost of tow train is applicable, which also achieves the wide application in many other literatures (Emde *et al.*, 2014). Because the number of deliveries is predetermined, the only outstanding problem is the inventory cost. As a result, the objective of this paper is to minimize total line-side inventory for all stations over the planning horizon. In the light of the assumptions and notation definitions above, the scheduling model can be formulated as the following objective function subject to feasibility constraints in equations (2)–(5).

Objective function:

$$\text{Minimize} \quad \sum_{c=1}^C \sum_{s=1}^{|S|} IL(s, c) \quad (1)$$

Subject to:

$$\sum_{k=1}^K b_{ks} + IL(s, 0) = \sum_{c=1}^C d_{sc} \quad \forall s \in S \quad (2)$$

$$0 \leq IL(s, c) \leq B_s \quad \forall s \in S, \quad c = 1, 2, \dots, C \quad (3)$$

$$0 < \sum_{s=1}^{|S|} b_{ks} \leq A \quad \forall k = 1, 2, \dots, K \quad (4)$$

$$t_k + R + \delta = t_{k+1} \text{ and } t_1 = 1, t_k \leq C \quad \forall k = 1, 2, \dots, K \quad (5)$$

where:

$$O(c, s) = \sum_{k=1}^K b_{ks} \mu_1(t_k + \lceil p_s \rceil, c) \quad \forall s \in S, c = 1, 2, \dots, C \quad (6)$$

$$D(c, s) = \sum_{c'}^c d_{sc'} \quad \forall s \in S, c = 1, 2, \dots, C \quad (7)$$

$$IL(s, c) = IL(s, 0) + O(c, s) - D(c, s) \quad \forall s \in S, c = 1, 2, \dots, C \quad (8)$$

Objective function (1) aims at minimizing total line-side inventory for all stations over the planning horizon. Equation (2) ensures that the materials delivered meet the total demand of workstations. Constraint [equation (3)] indicates that parts must be conveyed to the station timely to avoid stock-out, and the inventory for each station does not exceed capacity limits. Constraint [equation (4)] makes sure that part quantity within each tour would not violate the capacity constraint of tow trains. Equation (5) shows that the departure time of a tow train is cyclical and accords with the no-wait policy. Additionally, the cumulated number of parts delivered to station s and required at station s in work cycle c is given by equations (6) and (7), respectively. The mathematical expression of inventory level of the workstation is defined in equation (8).

3.3 Theorems of the research problem

This subsection mainly aims to enlighten the solution mechanism and eliminate some non-improvement-added adjustments without comprehensively evaluating the achieved solutions. Consequently, the following theorems are proposed on the basis of analysis of the problem domain.

Lemma 1: Without the consideration of capacity limitations of the inventory of workstations, any feasible solution should satisfy the following constraint:

$$\sum_{s=1}^{|S|} IL_{sk} \geq U_k, \quad \forall k = 1, 2, \dots, K$$

where IL_{sk} denotes the inventory of workstation s after the delivery k , U_k means the minimum inventory of the workstation that meets the condition of no stock-out after the delivery k .

Proof: Based on the definition of IL_{sk} , U_k mentioned above, we can conclude that without the consideration of capacity limit of line-side inventory for workstations, when $\sum_{s=1}^{|S|} IL_{sk} \geq U_k$, it meets no shortage constraint, and any feasible solution should satisfy the constraint, thus any feasible solution of the problem meets $\sum_{s=1}^{|S|} IL_{sk} \geq U_k$.

Theorem 1: Without the consideration of capacity limit of line-side inventory for workstations, the lower bound of the scheduling problem $LB(f_{sum})$ is given by:

$$LB(f_{sum}) = \sum_{k=1}^K \sum_{s=1}^{|S|} \sum_{c=t_k + \lceil p_s \rceil}^{t_{k+1} + \lceil p_s \rceil - 1} \left(\sum_{c'=t_k + \lceil p_s \rceil}^{t_{k+1} + \lceil p_s \rceil - 1} d_{sc'} - \sum_{c'=t_k + \lceil p_s \rceil}^c d_{sc'} \right) + \sum_{k=1}^K U_k$$

where d_{sk} represents material required at station s on tour k and ED_k denotes additional material requirements on tour k .

Proof: First, it is essential to clarify the implication of variables such as d_{sk} , ED_k , U_k , IL_{sk} :

$$d_{sk} : d_{sk} = \begin{cases} \sum_{c=t_k + \lceil p_s \rceil}^{t_{k+1} + \lceil p_s \rceil - 1} d_{sc}, & k = 1, 2, \dots, K \\ \sum_{c=t_k + \lceil p_s \rceil}^C d_{sc}, & k = K \end{cases}$$

$$ED_k : ED_k = \sum_{s=1}^{|S|} d_{sk} - A, \quad \forall k = 1, 2, \dots, K$$

If the result is positive, it not only reflects the loading capacity of tow trains but also indicates that materials fail to be delivered in time on tour k and therefore should be conveyed ahead of time.

$$U_k : U_k = \max\{0; U_{k+1} + ED_{k+1}\}, \quad \forall k = 1, 2, \dots, K$$

where $U_k = 0$.

$$IL_{sk} : IL_{sk} = \sum_{k'}^k b_{sk'} - \sum_{k'}^k d_{sk'}, \quad \forall k = 1, 2, \dots, K, s \in S$$

$$t_{K+1} : \text{Let } t_{K+1} = C$$

be the fictitious departure time for the sake of subsequent descriptions. The material requirement d_{sc} exceeding the work cycle C is 0.

Then, the significance of total inventory f_{sum} is under thorough analysis. It consists of two parts: one is inventory f_1 emerging before the next delivery, which exists inevitably, and the other is inventory f_2 due to the delivery in advance. Let

$$\varnothing_k = \sum_{s=1}^{|S|} \sum_{c=t_k + \lceil p_s \rceil}^{t_{k+1} + \lceil p_s \rceil - 1} \left(\sum_{c'=t_k + \lceil p_s \rceil}^{t_{k+1} + \lceil p_s \rceil - 1} d_{sc'} - \sum_{c'=t_k + \lceil p_s \rceil}^c d_{sc'} \right)$$

be the inventory emerging between tour k and tour $k+1$, hence

$f_1 = \sum_{k=1}^K \varnothing_k$ is a constant and $f_2 = \sum_{k=1}^K \sum_{s=1}^{|S|} IL_{sk}$ is a variable, so to minimize f_{sum} is to minimize f_2 . According to Lemma 1, any feasible solution satisfies the constraint $\sum_{s=1}^{|S|} IL_{sk} \geq U_k$, thus $f_2 = \sum_{k=1}^K \sum_{s=1}^{|S|} IL_{sk} \geq \sum_{k=1}^K U_k$, $f_{sum} \geq f_1 + \sum_{k=1}^K U_k$, namely, the lower bound of f_{sum} is:

$$LB(f_{sum}) = \sum_{k=1}^K \sum_{s=1}^{|S|} \sum_{c=t_k + \lceil p_s \rceil}^{t_{k+1} + \lceil p_s \rceil - 1} \left(\sum_{c'=t_k + \lceil p_s \rceil}^{t_{k+1} + \lceil p_s \rceil - 1} d_{sc'} - \sum_{c'=t_k + \lceil p_s \rceil}^c d_{sc'} \right) + \sum_{k=1}^K U_k$$

Theorem 1 is proven.

Theorem 2: If $\varphi_1, \varphi_2 \in \{1, 2, \dots, K\}$ ($\varphi_1 < \varphi_2$) denote two delivery jobs of the integrated material handling scheme Ω , then

when $\forall s \in S$ and $\forall k \in \{\varphi_1 + 1, \dots, \varphi_2 - 1\}$ satisfies $\begin{cases} b_{\varphi_1 s} \neq 0 \\ b_{\varphi_2 s} \neq 0, \\ b_{ks} = 0 \end{cases}$

$IL(s, c)$ which decreases progressively in the interval of $[t_{\varphi_1} + \lceil p_s \rceil, t_{\varphi_2} + \lceil p_s \rceil - 1]$ obtains the maximum and minimum values at both ends. Theorem 2 is clearly established and the proof is omitted.

Based on Theorem 2, it can be concluded that when delivery jobs of scheme Ω denoted by φ_1, φ_2 can satisfy the above conditions, the inventory of station s at time $t = t_{\varphi_1} + \lceil p_s \rceil$ might violate the capacity restriction of line-side inventory, and a stock-out may take place at station s at time $t = t_{\varphi_2} + p_s - 1$. Accordingly, the feasibility of solutions can be judged by the construction of the maximum and minimum inventory set which are denoted by $\Gamma_{\max}, \Gamma_{\min}$ and can be calculated by the following equations (9) and (10).

$$\Gamma_{\max} = \{IL(s, t) \mid t = \min(t_k + \lceil p_s \rceil, C), k \in \{1, 2, \dots, K\}\} \quad (9)$$

$$\Gamma_{\min} = \{IL(s, t) \mid t = \min(t_k + \lceil p_s \rceil - 1, C), k \in \{1, 2, \dots, K\}\} \quad (10)$$

If $\exists \gamma^b > 0$ ($\gamma^b \in \Gamma_{\min}$), there exists a stock-out in the assembly line, consequently the scheme Ω is an infeasible solution. If $\exists \gamma^a > B_s$ ($\gamma^a \in \Gamma_{\min}$), it indicates that the capacity constraint of line-side workstations is violated, and the scheme is also an infeasible solution.

Theorem 3: If the minimum number of bins delivered to station s on tour φ_1 is denoted by $b_{\varphi_1 s}^{\min}$ and if $\varphi_1, \varphi_2 \in \{1, 2, \dots, K\}$ ($\varphi_1 < \varphi_2$) denote two delivery jobs of the integrated material handling scheme Ω , then when $\forall s \in S$ and $\forall k \in \{\varphi_1 + 1, \dots, \varphi_2 - 1\}$ satisfies

$$\begin{cases} b_{\varphi_1 s} \neq 0 \\ b_{\varphi_2 s} \neq 0, \\ b_{ks} = 0 \end{cases} \quad \text{there exists}$$

$$b_{\varphi_1 s}^{\min} = \sum_{j=1}^{t_{\varphi_2} + \lceil p_s \rceil - 1} d_{sj} - IL(s, 0) - \sum_{i=1}^{\varphi_1 - 1} b_{is}.$$

Proof: If $\forall k \in \{\varphi_1 + 1, \dots, \varphi_2 - 1\}$ satisfies $\begin{cases} b_{\varphi_1 s} \neq 0 \\ b_{\varphi_2 s} \neq 0, \\ b_{ks} = 0 \end{cases}$, then it

can be concluded that φ_1 and φ_2 are two delivery jobs adjacent to station s , thus $b_{\varphi_1 s}^{\min}$ which is the inventory at station s at time

$(t_{\varphi_2} + \lceil p_s \rceil - 1)$ on tour φ_2 is just 0, that is to say,

$$IL(s, t_{\varphi_2} + \lceil p_s \rceil - 1) = 0.$$

Theorem 3 is proven.

4. NSICSA design

As discussed in the above section, the problem in this article falling under the category of NP-hard brings significant difficulty in obtaining optimal solutions for exact algorithms. Consequently, ICSCA is introduced as the main framework for the optimization algorithm. The ICSCA is distinguished for its excellent data and information processing but comes short on local search processes. To ameliorate the capability of the proposed algorithm, NS is incorporated with ICSCA to improve

the exploitation ability in local search. And in the meanwhile, SA is integrated into the hybrid algorithm to avoid local equilibrium when the algorithm evolves to a certain extend. Accordingly, in this paper, an ICSCA combined with the NS and SA operators which the authors call the NSICSCA is developed, possessing the global search ability of ICSCA, the ability of getting rid of local optimum of SA and the deep search ability of NS to get better solutions. The detailed descriptions of NSICSCA are presented as follows.

4.1 Encoding schemes

Two of the key procedures of the immune algorithm are the antibody identification and the antibody coding. According to the problem characteristics, an antibody denotes a material handling scheme, namely, a solution, and a coding scheme example is illustrated in Figure 2, where $K = 3, |S| = 8$. The scheme has $|S|$ columns and K rows, in which each column represents the workstation $s (s \in S)$, each row represents the tour $k (k = 1, 2, \dots, K)$, and b_{ks} denotes the element x_{ks} , which means the bins delivered to the workstation s on tour k . The sum of each row of the scheme should be less than the loading capacity of the tow train to satisfy the equation (4). In addition, the sum of each column of the scheme should be equal to the total demand of workstations to satisfy the equation (2).

4.2 Population initialization

The initialization of the antibody is crucial to the convergence of algorithms and the quality of solutions. Hence, an approach based on chaos is proposed to initialize the antibody. The chaotic iteration parameter CI and the population size N_{pop} are given, and the main procedures are illustrated as follows:

Step 1: Let $AP^0 \leftarrow \emptyset, i = 1, j = 1$.

Step 2: Generate $K \times |S|$ different random numbers $\Delta_{mn}^0 (m = 1, 2, \dots, K; n = 1, 2, \dots, |S|) \in [0, 1]$.

Step 3: $\forall \Delta_{mn}^i (m = 1, 2, \dots, K; n = 1, 2, \dots, |S|)$, the following chaotic operations are performed until the generation of Δ_{mn}^{CI} :

$$\begin{aligned} &\text{for } i = 1: CI \\ &\quad \Delta_{mn}^i = \sin(\pi \cdot \Delta_{mn}^{i-1}) \\ &\text{end for} \end{aligned}$$

Step 4: $LB_{ks} = \min\{A - \sum_{s'=1}^s b_{ks'}, \sum_{k'=1}^k d_{sk'} - \sum_{k'=1}^{k-1} b_{k's}\}$

$$\varepsilon = \min\{A - \sum_{s'=1}^s b_{ks'} - LB_{ks}, \sum_{k=1}^K d_{sk} - \sum_{k=1}^{k-1} b_{k's} - \sum_{k'=1}^k d_{sk'}\},$$

then $b_{ks} = LB_{ks} + \text{rand}(\Delta_{mn}^{CI} \cdot \varepsilon)$, until the antibody Ω_1 is generated.

Repeat the steps above N_{pop} times to generate the initial antibody group $AP^0 = \{\Omega_1, \Omega_2, \dots, \Omega_{N_{pop}}\}$.

4.3 Termination condition judgement

If the number of iterations reaches the specified upper limit $MAXGEN$, the solution result is output; otherwise, proceed to step 4.4.

Figure 2 Coding scheme example

1	0	3	1	2	1	0	3
1	1	0	1	0	1	3	2
2	3	1	0	2	0	1	1

4.4 Construction of affinity function

Owing to the omission of considering the inventory shortage and capacity constraints of workstations in the coding scheme, infeasible solutions may exist in the antibody population if the line-side inventory level meets the condition proposed in Theorem 2. Under the circumstance, the penalty value should be introduced to the objective function when it comes to the affinity evaluation.

The performance of antibodies is evaluated by its affinity. Because the objective of the problem is to minimize total line-side inventory for all stations over the planning horizon, the affinity function for an antibody Ω is calculated as follows:

$$\text{Affinity}(\Omega) = \frac{1}{IL(\Omega) + \text{pena}(\Omega)} \quad (11)$$

where $IL(\Omega)$ is the corresponding objective function of antibody Ω , and $\text{pena}(\Omega)$ denotes the penalty function. Based on Theorem 2, the $\Gamma_{\max}$ and $\Gamma_{\min}$ of scheme Ω can be constructed. Furthermore, if $\exists \gamma^\alpha > B_s(\gamma^\alpha \in \Gamma_{\max})$ or $\exists \gamma^\alpha > B_s(\gamma^\alpha \in \Gamma_{\min})$, which means that Ω is an unfeasible solution, the corresponding penalty measures are forced to take place in the affinity evaluation. The penalty function $\text{pena}(\Omega)$ is defined as follows:

$$\text{pena}(\Omega) = - \left(\sum_{i=1}^{|\Gamma_{\max}|} \lambda_1 \cdot \max(\gamma_i^\alpha - B_s, 0) + \sum_{j=1}^{|\Gamma_{\min}|} \lambda_2 \cdot \min(\gamma_j^b, 0) \right) \quad (12)$$

where λ_1 and λ_2 are the penalty coefficients for the stock-out of workstations and the stock exceeding the inventory capacity of workstations, respectively. For the convenience in programming, the performance of antibodies is directly evaluated by total inventory which is denoted by IL_{sum} where $IL_{\text{sum}} = IL(\Omega) + \text{pena}(\Omega)$.

4.5 Clonal selection operation

The affinity of each antibody is evaluated, and then the top N_c antibodies are selected and stored in the memory bank in the descending order of size, and the position of which in the sequence is denoted by r . They are then cloned separately, and the number of clones is $(N_c - r + 1)$, thus the total number of antibodies after the completion of cloning is $N_c(N_c + 1)/2$. The affinity of the antibody population is improved through the cloning operation. An illustrative example is depicted in Figure 3.

4.6 Neighborhood search

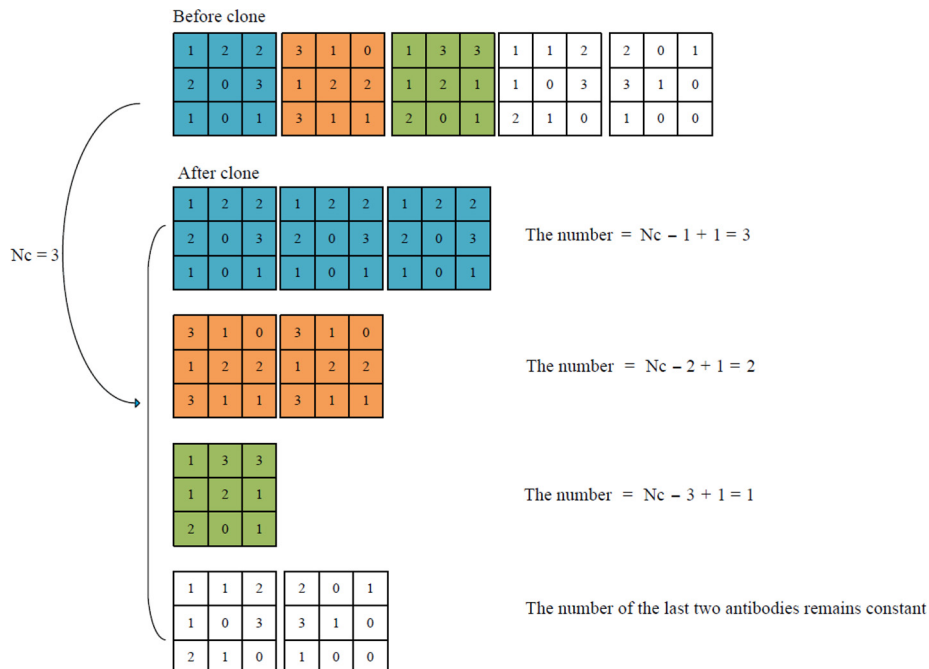
Considering the significant influence of the current optimum solution on the convergence of the algorithm, the NS operator is used to perform deep search on the individuals in the current memory bank to find the “critical chain” of the antibody and the corresponding “key station” denoted by s^* . The operator can make the solution develop in a better direction without destroying the memory bank. s^* is given by:

$$s^* = \arg \max IL(s, c) \quad (s \in S, c = 1, 2, \dots, C) \quad (13)$$

For the antibody Ω_{M_i} , $\Omega \leftarrow \Omega_{M_i}$, s^* is obtained by the formula (13), the total inventory of $\sum_{c=1}^C IL(s^*, c)$ will be the minimum if satisfying the following constraints:

$$\begin{cases} b_{ks^*} \leq A - \sum_{j \in S \text{ and } j \neq s^*} b_{kj} \\ \sum_{k=1}^K b_{ks^*} = \sum_{c=1}^C d_{s^*c} \end{cases}, \quad \forall k = 1, 2, \dots, K \quad (14)$$

Figure 3 Clonal operation



4.7 Hyper-mutation operations

Unlike GAs, immune algorithms mutate all antibodies in a cloned population according to their affinity values (Abdollahpour and Rezaeian, 2015). Given the coefficient of mutation $0 < M_1 < M_2$, the antibody is subjected to the following mutations:

Step 1: Given the scheme Ω , $\forall k = 1, 2, \dots, K, s \in S$, $b_{ks}^{\min}$ is calculated according to Theorem 3. If $b_{ks}^{\min} < 0$, which indicates that the line-side inventory of s is in high level, so the first $(k-1)$ times of delivery can meet the demand. Therefore, the delivery on tour $k' (k' > k)$ is randomly selected, and there exists $b_{k's} \leftarrow b_{k's} + b_{ks}$, $b_{ks} \leftarrow 0$. If $b_{ks}^{\min} > A$, then the inventory of s is in low level, hence the delivery needs to be arranged in advance. The previous delivered bins are given by $b_{k-1,s} \leftarrow b_{k-1,s} + \text{rand}(0,1) \cdot \Delta$, where $\Delta = \min\left\{A - \sum_{s=1}^{|S|} b_{k-1,s}, b_{ks}\right\}$, which constitutes Ω' , $\Omega \leftarrow \Omega'$.

Step 2: If $r < M_1$, then go to step3; else if $M_1 < r < M_2$, then go to step 4. Step 3: Choose

$\forall s \in S$, if $\exists k_1, k_2 \in \{1, 2, \dots, K\}$, satisfying $b_{k_2s} \leq A - \sum_{j=1}^{|S|} b_{k_1j}$, then $b_{k_1s} \leftarrow b_{k_1s} + b_{k_2s}$, $b_{k_2s} \leftarrow 0$, which constitutes Ω' , $\Omega \leftarrow \Omega'$, and then go to step5. The procedure is performed as shown in Figure 4(a).

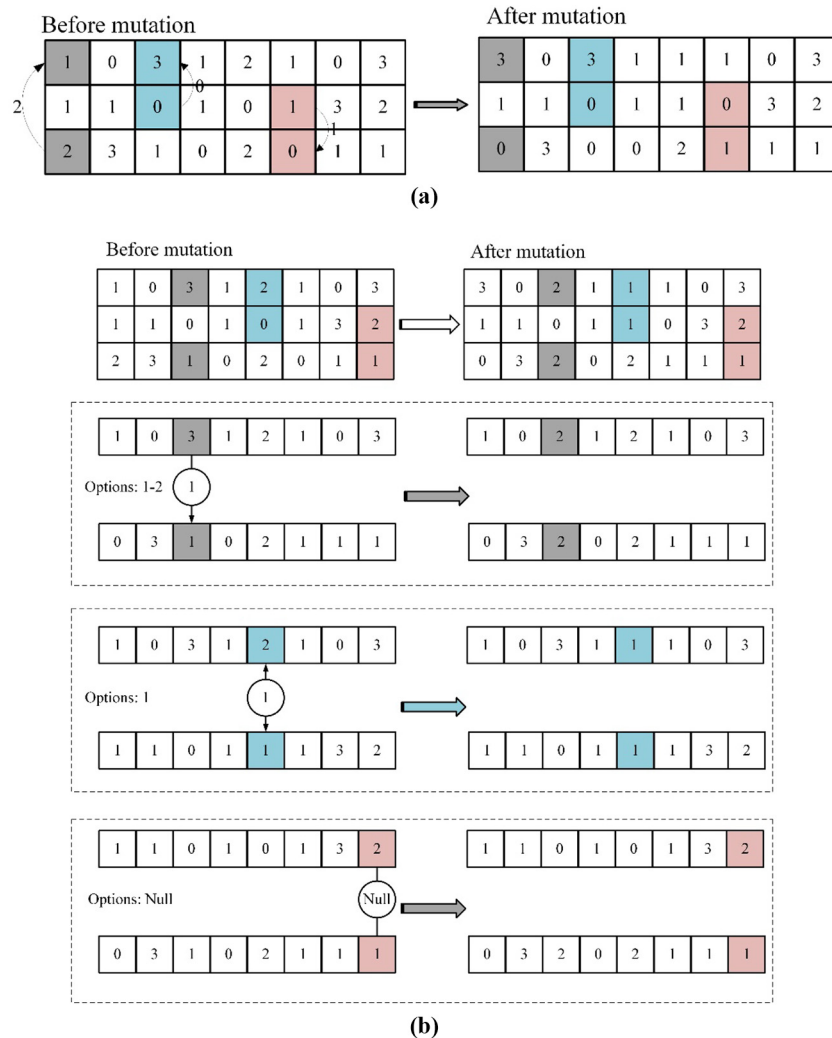
Step 4: Choose $\forall s \in S$, if $\exists k, k' \in \{1, 2, \dots, K\}$, satisfying $b_{ks} > 1$, then $\forall b' \in \{1, 2, \dots, b_{ks} - 1\}$ and $b' \leq A - \sum_{j=1}^{|S|} b_{k'j}$, $b_{ks} \leftarrow b_{ks} - b'$, $b_{k's} \leftarrow b_{k's} + b'$. Go to step5. The procedure is carried out as shown in Figure 4(b).

Step 5: Antibody mutation ends.

4.8 Simulated annealing operations

The immune clone selection algorithm may fall into local equilibrium after evolving to a certain extent. Consequently, when the algorithm runs to the later stage, the SA operation is introduced to help the algorithm escape from the local equilibrium state. Finally, update the population and go to step 4.3. The pseudo codes of SA operations are shown in Procedure of SA:

Figure 4 Procedure of hyper-mutation operations



Notes: (a) The mutation process of step 3; (b) The mutation process of step 4

begin

Initialize the initial temperature $t = t_0$, the initial state $s = s_0$, $k = 0$

Repeat

Repeat

Generate new state $S_j = \text{Generate}(s)$;

while sampling stability criteria is not met **do**

if $\min\{1, \exp[-c(s_j - c(s)/t_k)]\} \geq \text{random}(0, 1)$

$s = s_j$

end

end

while algorithm termination criteria are not met **do**

Annealing $t_{k+1} = \text{update}(t_k)$

$k \leftarrow k + 1$

end

Input the searching results

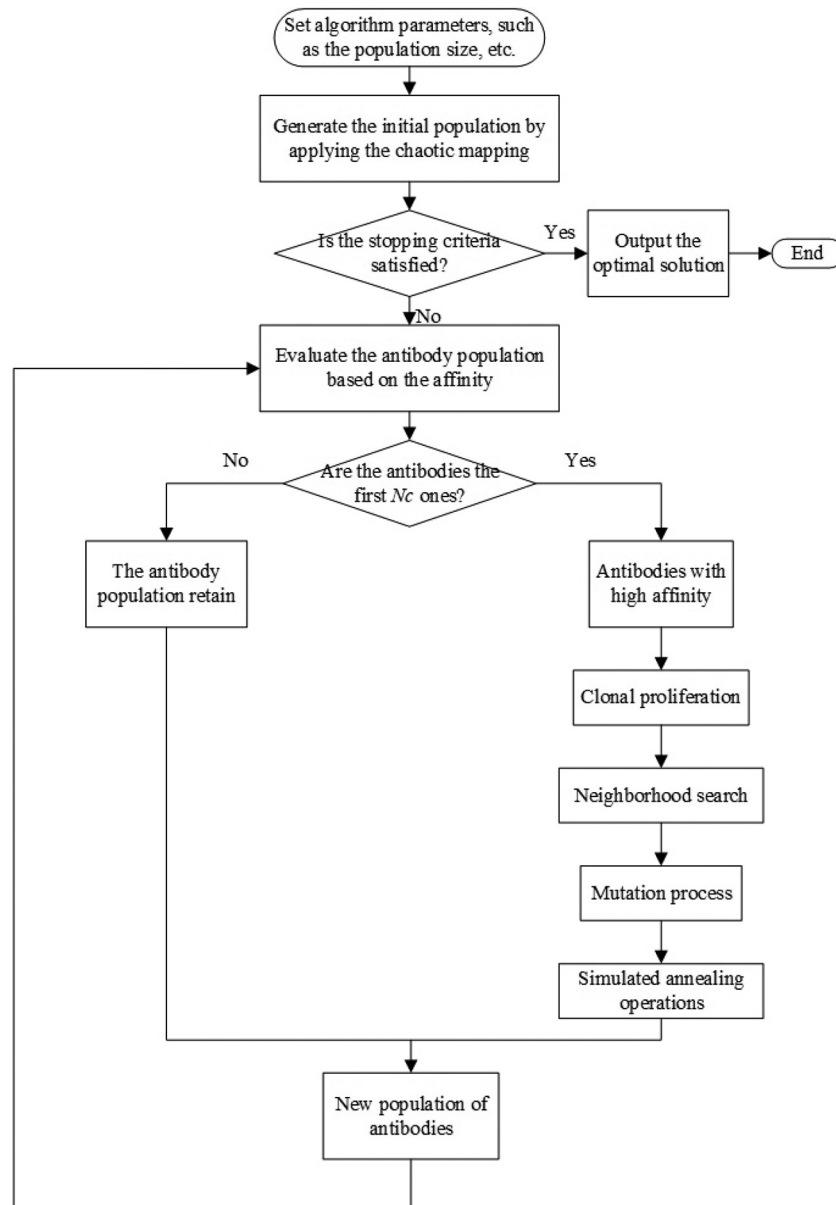
4.9 Overall procedure of NSICSA algorithm

Based on detailed descriptions in subsections 4.1–4.8, the overall procedure of NSICSA algorithm is depicted in Figure 5.

5. Computational experiments

To assess the performance of the NSICSA proposed in this article, we implemented our algorithms in MATLAB(R2016a) to solve a set of test problem instances with distinct characteristics on a PC equipped with an Intel Core i5 1.8 GHz CPU, 8 GB RAM and Windows 10 operating system. The generation of test instances is detailed in section 5.1. Section 5.2 shows the comparison among the proposed NSICSA and two benchmark algorithms by 60 computational tests with boxplots and convergence curves. A series of analyses on computational complexity and the effect of inventory capacity

Figure 5 Procedure of NSICSA algorithm



on inventory levels are carried out in sections 5.3 and 5.4. Finally, studies on managerial applications are discussed in section 5.5 to get managerial insights.

5.1 Experimental settings

5.1.1 Instances generation.

To verify the effectiveness of the NSICSA, the test examples $(|S|, C)$ are constructed as follows referring to Boysen and Emde (2014). Three problem sizes have been considered: small (fewer than 25 workstations and 40 work cycles, namely, $|S| \leq 25, C \leq 40$), medium (between 40 and 80 work cycles, 30 and 50 workstations, namely, $30 \leq |S| \leq 50, 40 \leq C \leq 80$) and large (more than 50 workstations and 80 work cycles, namely, $|S| > 50, C > 80$). The round-trip time of each tow train is $R = \lceil p_{|s|} + \text{rand}(0.05, 0.2) \rceil$, and the driving time from the supermarket to the first workstation is $p_1 = \text{rand}(0.05, 0.2)$ whilst the driving time from the supermarket to the other workstations is $p_s = p_{s-1} + \text{rand}(0.05, 0.2), \forall s \in \{2, 3, \dots, |S|\}$, and the loading capacity of the tow train is set to $A = (\sum_{s=1}^{|S|} \sum_{c=1}^C d_{sc}) / K + 1$. In addition, $d_{sc} = 1$ or 0, and the probability of $d_{sc} = 1$ is 45 per cent to 50 per cent. Table I summarizes the other parameter

settings for the instance generation where each instance is labeled in the form of $(|S|, C)$. For example, instance (5,27) denotes the scheduling problem with 5 workstations and 27 work cycles.

Figure 6 Convergence curves of different algorithms

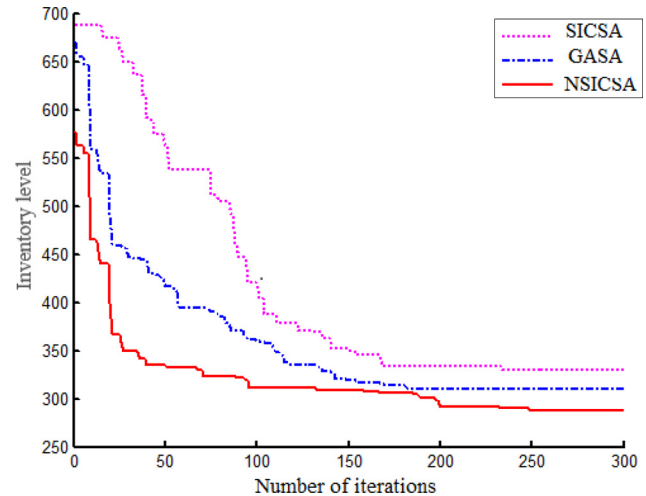


Table I Parameter settings for instance generation

Notations	Explanation	Small	Medium	Large
$ S $	Number of workstations	5/10/15/20/25	30/35/40/45/50	55/60/65/70/75
C	Work cycles	9/18/27/36	45/54/63/72	81/90/99/108
δ	Material replenishment time in the supermarket	4	8	12
$IL(s,0)$	Initial inventory of each workstation	1	1	1
B_s	Inventory capacity of each workstation	3	3	3

Table II Parameters of metaheuristic algorithms

Algorithms	Parameter	Low	Value Middle	High
NSICSA	Number of population	55	80	105
	Starting generation of SA	300	350	400
	Coefficient of mutation	(0.4,0.8)	(0.4,0.8)	(0.4,0.8)
	Initial temperature	1,000	1,250	1,500
	Cooling rate of temperature	0.9	0.94	0.98
	Maximum number of SA	20	25	30
GASA	Maximum generation	1,000	1,500	2,000
	Number of population	55	80	105
	Starting generation of SA	300	350	400
	Crossover probability	0.6	0.65	0.7
	Mutation probability	0.4	0.45	0.5
	Initial temperature	1,000	1,250	1,500
SICSA	Cooling rate of temperature	0.9	0.94	0.98
	Maximum number of SA	20	25	30
	Maximum generation	1,000	1,500	2,000
	Number of population	55	80	105
	Coefficient of mutation	(0.4,0.8)	(0.4,0.8)	(0.4,0.8)
	Maximum generation	1,000	1,500	2,000

5.1.2 Parameters setting

We compared our proposed NSICSA to the GASA (Rao *et al.*, 2013) and the standard immune clonal selection algorithm (SICSA) and executed each algorithm 10 times with different random seeds. The algorithms are selected for the following

two reasons: First, the structure and framework of GASA and SICSA are similar to our proposed algorithm; thus, they can be comparable. Second, the basic algorithm of GASA, namely, GA and SICSA are widely used as benchmark algorithms in many articles to provide baseline results when evaluating

Table III Computational results of small-scale instances

No.	(S ,C)	LB	NSICSA			GASA			SICSA	
			IL_{sum}	PD(%)	IL_{sum}	PD(%)	GAP(%)	IL_{sum}	PD(%)	GAP(%)
1	(5,9)	22	22	0	22	0	0	22	0	0
2	(5,18)	43	43	0	43	0	0	43	0	0
3	(5,27)	76	76	0	76	0	0	79	3.95	3
4	(5,36)	98	99	1.02	101	3.06	2.02	103	5.10	4.04
5	(10,9)	45	45	0	45	0	0	45	0	0
6	(10,18)	81	81	0	83	2.47	2.47	84	3.70	3.70
7	(10,27)	149	154	3.36	156	4.70	1.30	157	5.37	1.95
8	(10,36)	184	192	4.35	195	5.98	1.56	198	7.61	3.13
9	(15,9)	74	74	0	74	0	0	75	1.35	1.35
10	(15,18)	156	162	3.84	165	5.77	1.85	166	6.41	2.47
11	(15,27)	214	225	5.14	228	6.54	1.33	231	7.94	2.67
12	(15,36)	278	293	5.40	296	6.48	1.02	302	8.63	3.07
13	(20,9)	102	104	1.96	106	3.92	1.92	108	5.88	3.85
14	(20,18)	192	201	4.67	204	6.25	1.49	207	7.81	2.99
15	(20,27)	285	301	5.61	304	6.67	1.00	310	8.77	2.99
16	(20,36)	364	387	6.31	395	8.52	2.07	397	9.06	2.58
17	(25,9)	137	141	2.92	143	4.38	1.42	145	5.84	2.84
18	(25,18)	232	244	5.17	247	6.47	1.23	251	8.19	2.87
19	(25,27)	359	385	7.24	393	9.47	2.08	391	8.91	1.56
20	(25,36)	458	495	8.08	506	10.48	2.22	511	11.57	3.23
Average	–	177.5	186.2	4.93	189.1	6.57	1.56	191.3	7.78	2.71

Table IV Computational results of medium-scale instances

No.	(S ,C)	LB	NSICSA			GASA			SICSA	
			IL_{sum}	PD(%)	IL_{sum}	PD(%)	GAP(%)	IL_{sum}	PD(%)	GAP(%)
1	(30,45)	684	748	9.36	761	11.26	1.74	769	12.43	2.81
2	(30,54)	790	873	10.51	892	12.9	2.18	897	13.54	2.75
3	(30,63)	926	1,029	11.12	1,049	13.28	1.94	1,056	14.04	2.62
4	(30,72)	1,049	1,175	12.01	1,201	14.49	2.21	1,216	15.92	3.49
5	(35,45)	782	860	9.97	876	12.02	1.86	883	12.92	2.67
6	(35,54)	938	1,046	11.51	1,067	13.75	2.00	1,078	14.93	3.06
7	(35,63)	1,097	1,231	12.22	1,261	14.95	2.44	1,274	16.13	3.49
8	(35,72)	1,225	1,389	13.39	1,416	15.59	1.94	1,427	16.49	2.74
9	(40,45)	891	989	11.00	1,008	13.13	1.92	1,014	13.80	2.53
10	(40,54)	1,047	1,172	11.94	1,207	15.28	2.99	1,217	16.24	3.84
11	(40,63)	788	868	10.15	882	11.93	1.61	887	12.56	21.89
12	(40,72)	1397	1,599	14.46	1,625	16.32	1.62	1,658	18.68	3.69
13	(45,45)	1,009	1,127	11.69	1,156	14.57	2.57	1,164	15.36	3.28
14	(45,54)	1,186	1,334	12.48	1,369	15.43	2.62	1,381	16.44	3.52
15	(45,63)	1,403	1,614	15.04	1,648	17.46	2.11	1,674	19.32	3.72
16	(45,72)	1,572	1,817	15.59	1,868	18.84	2.81	1,894	20.48	4.24
17	(50,45)	1,127	1,265	12.24	1,296	15.00	2.45	1,309	16.15	3.48
18	(50,54)	1,304	1,489	14.19	1,511	15.87	1.48	1,537	17.87	3.22
19	(50,63)	1,553	1,791	15.33	1,835	18.16	2.46	1,861	19.83	3.91
20	(50,72)	1,741	2,051	17.81	2,107	21.02	2.73	2,194	26.02	6.97
Average	–	1,125.5	1,273.4	13.14	1301.8	15.66	2.23	1,319.5	17.24	3.62

Table V Computational results of large-scale instances

No.	(S ,C)	LB	NSICSA			GASA			SICSA	
			IL_{sum}	PD(%)	IL_{sum}	PD(%)	GAP(%)	IL_{sum}	PD(%)	GAP(%)
1	(55,81)	2,188	2,591	18.42	2,692	23.03	3.9	2,807	28.29	8.34
2	(55,90)	2,395	2,862	19.50	3,009	25.64	5.14	3,161	31.98	10.44
3	(55,99)	2,684	3,248	21.01	3,425	27.61	5.45	3,570	33.01	9.91
4	(55,108)	2,867	3,453	20.44	3,681	28.39	6.6	3,872	35.05	12.13
5	(60,81)	2,376	2,842	19.61	2,960	24.58	4.15	3,089	30.01	8.69
6	(60,90)	2,609	3,134	20.12	3,315	27.1	5.78	3,497	34.04	11.58
7	(60,99)	2,915	3,527	20.99	3,789	29.98	7.43	3,944	35.3	11.82
8	(60,108)	3,112	3,796	21.98	4,273	37.31	12.57	4,456	43.19	17.39
9	(65,81)	2,583	3,100	20.02	3,354	29.85	8.19	3,659	41.66	18.03
10	(65,90)	2,817	3,386	20.20	3,679	30.60	8.65	3,897	38.34	15.09
11	(65,99)	3,159	3,851	21.91	4,286	35.68	11.30	4,495	42.29	16.72
12	(65,108)	3,385	4,152	22.66	4,798	41.74	15.56	5,075	49.93	22.23
13	(70,81)	2,816	3,379	19.99	3,684	30.82	9.03	3,873	37.54	14.62
14	(70,90)	3,051	3,698	21.21	4,109	34.68	11.11	4,350	42.58	17.63
15	(70,99)	3,462	4,248	22.70	5,012	44.77	17.98	5,312	53.44	25.05
16	(70,108)	3,668	4,523	23.31	5,529	50.74	22.24	5,612	53.00	24.08
17	(75,81)	3,014	3,641	20.80	3,936	30.59	8.10	4,369	44.96	19.99
18	(75,90)	3,260	3,978	22.02	4,568	40.12	14.83	4,863	49.17	22.25
19	(75,99)	3,657	4,498	23.00	5,394	47.49	19.92	5,823	59.23	29.46
20	(75,108)	3,924	4,902	24.92	6,283	60.12	28.17	6,835	74.18	39.43
Average	–	2,997.1	3,640.5	21.47	4,088.8	36.43	12.32	4,328.0	44.40	18.89

brand-new optimization algorithms. The values of parameters are shown in Table II.

5.2 Computational results of NSICSA compared with GASA and SICSA

In this section, we will get the computational results of the NSICSA in comparison with the GASA and SICSA. First, we compare the convergence performance of the NSICSA, GASA and SICSA for one typical instance under the moderate problem scale, which is plotted in Figure 6. It is obvious that the convergence performance of the proposed NSICSA is significantly better than benchmark algorithms. The GASA and SICSA have a slower convergence rate and are easy to fall into local optima. It can also be seen that the NSICSA is superior to the GASA and SICSA in the quality of solutions. Solutions found by NSICSA appear to be of lower inventory level than the other two algorithms. Then we compare the three algorithms by 60 test cases which are divided into three groups on the basis of the problem scale (small, medium and large), each of which contains 20 cases.

The NSICSA and two benchmark algorithms are performed ten times for each case, and the average total inventory level IL_{sum} , the lower bound LB proposed in Theorem 1, the percentage deviation (PD) and the gap (GAP) are calculated. The PD and GAP are given by the following equations:

$$PD = \frac{IL_{sum} - LB}{LB} \times 100\% \quad (15)$$

$$GAP = \frac{IL_{sum}^{GASA} - IL_{sum}^{NSICSA}}{IL_{sum}^{NSICSA}} \times 100\% \quad (16)$$

$$GAP = \frac{IL_{sum}^{SICSA} - IL_{sum}^{NSICSA}}{IL_{sum}^{NSICSA}} \times 100\% \quad (17)$$

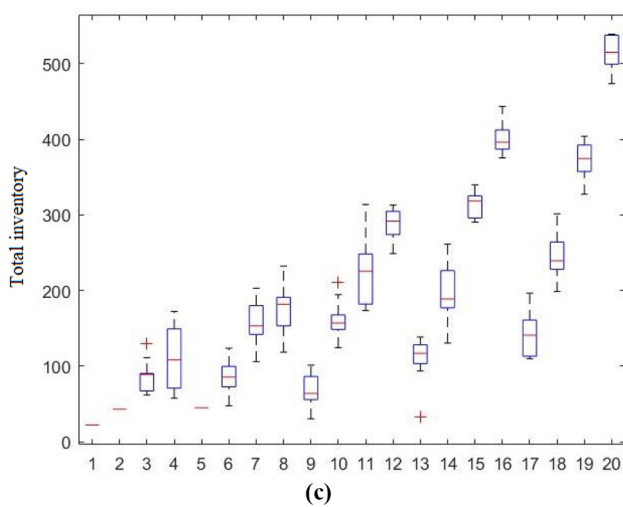
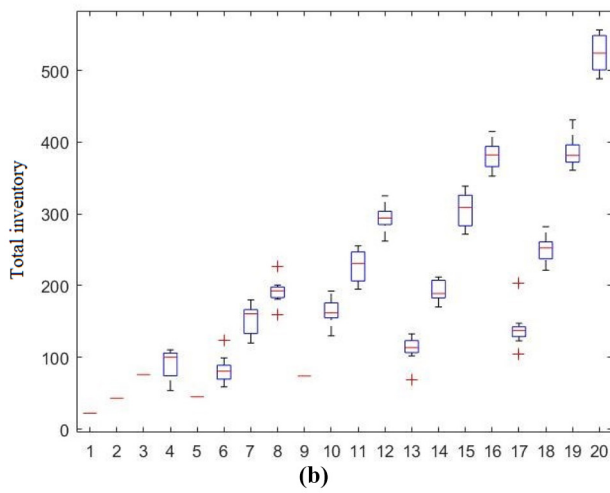
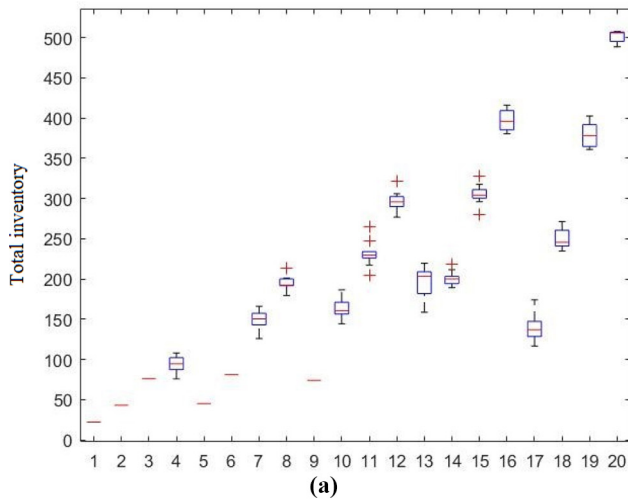
where IL_{sum}^{NSICSA} , IL_{sum}^{GASA} and IL_{sum}^{SICSA} denote the IL_{sum} of the NSICSA, GASA and SICSA respectively. Tables III–V summarize the computational results of different problem scales.

Moreover, to better visualize the performance of the proposed NSICSA and two benchmark algorithms in problems of different scales, the boxplot (Figures 7–9) is used, where the total inventory values of each algorithm generated in ten runs are shown.

According to the computational results, the following conclusions are summarized:

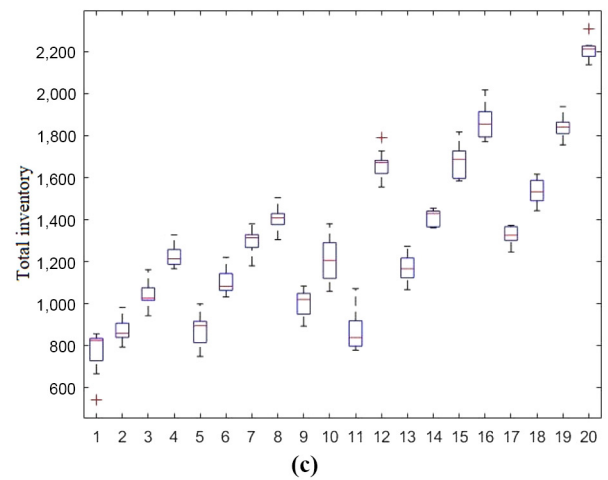
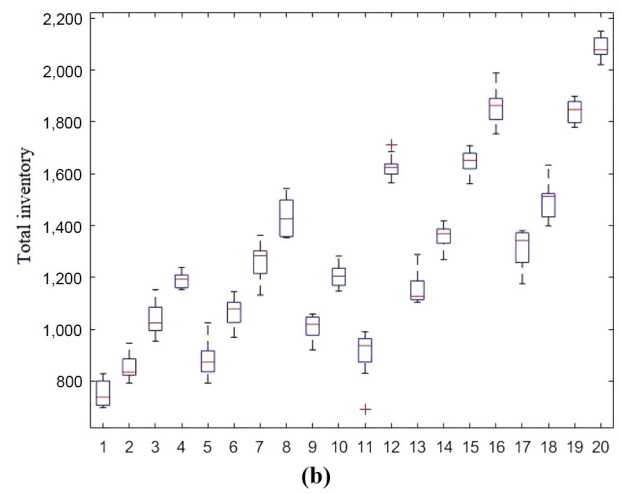
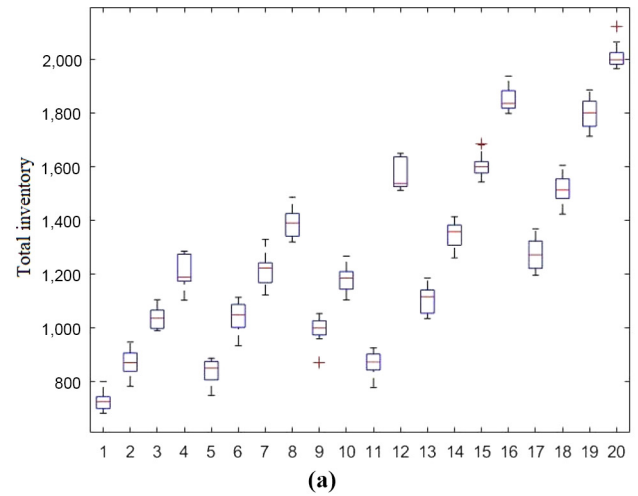
- With respect to the IL_{sum} column, the solutions obtained by the NSICSA are significantly lower in most cases compared with the GASA and SICSA, especially when the problem scale gets larger. It demonstrates that the solutions obtained by the NSICSA are of higher quality. In addition, this proves that the convergence performance of the NSICSA is more superior than the other two benchmark algorithms, which confirms the convergence speed shown in Figure 6.
- For the $PD(\%)$ column, it can be concluded that the $PD(\%)$ of NSICSA is always within 25 per cent, which indicates that the effectiveness validation of NSICSA is confirmed. In addition, compared with the other benchmark algorithms, the optimization performance of the proposed algorithm is significantly better. As the problem scale increases, the advantage is more obvious especially when the PD of the GASA and SICSA reach

Figure 7 Boxplots of total inventory level in the small-scale instance



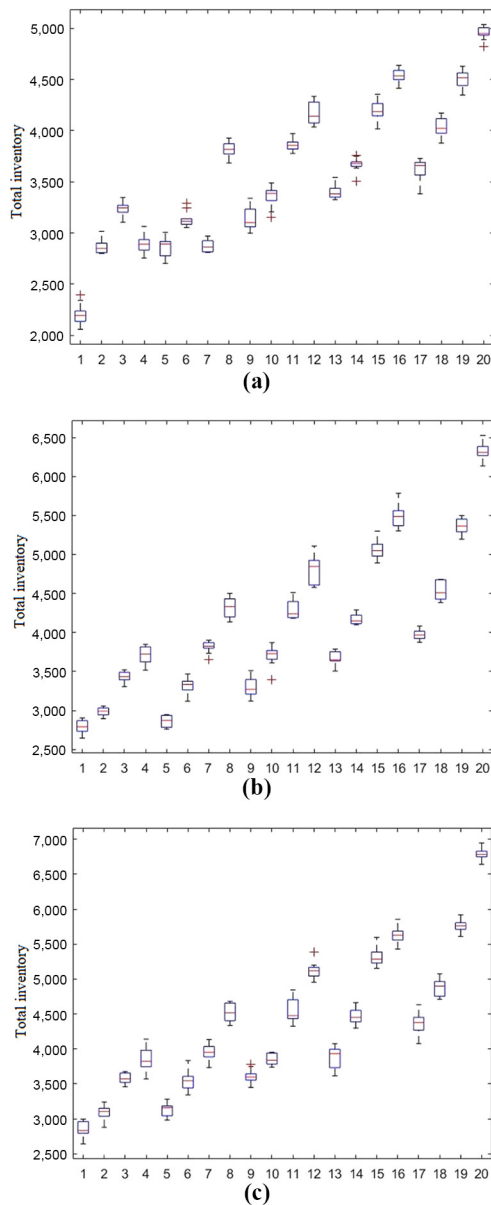
Notes: (a); (b); (c)

Figure 8 Boxplots of total inventory level in the medium-scale instance



Notes: (a); (b); (c)

Figure 9 Boxplots of total inventory level in large-scale instance



Notes: (a); (b) and (c)

60.12 and 74.18 per cent, respectively, in the largest scale instance.

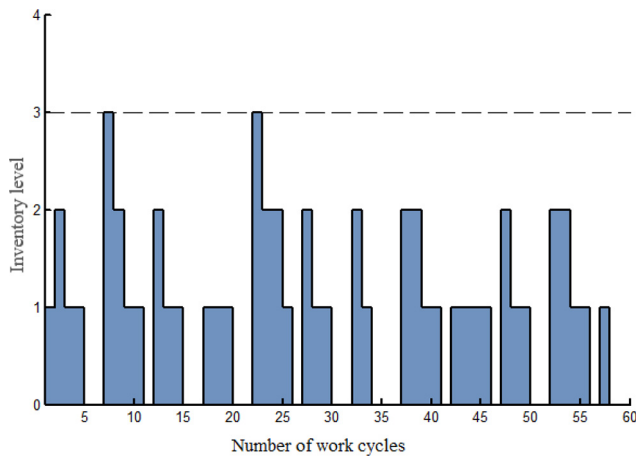
- For the *GAP*(%) column, we can see that the gap between two benchmark algorithms and the NSICSA is positive in most cases of every scale, and it gradually becomes apparent when the problem scale increases. It means that the results obtained by the NSICSA are significantly better than those of GASA and SICSA. Consequently, the computational results show that the NSICSA is capable of solving the scheduling and loading problem of tow trains more effectively.
- From the three boxplots of total inventory level in each scale case, it can be noticed that the NSICSA has a good performance because its computational results have better mean values with low variabilities in most cases. Therefore, it can be concluded that the NSCISA is more stable than the other two benchmark algorithms.
- With regard to the main results, solutions obtained by the NSICSA algorithm appear to be of lower inventory level than the other two algorithms. In addition, it is obvious that the distribution of the NSICSA is more concentrated than that of the GASA and SICSA, which indicates that the NSICSA is superior to other two algorithms in terms of stability. The inventory level remains low and stable even when the problem scale increases to the real-world scale that is extremely similar to the scale of the actual assembly line in application. Under this circumstance, the solution obtained by the NSICSA algorithm is of great managerial significance in cutting down the inventory cost, thus cutting down the total system cost including the material delivery cost and the inventory cost. Moreover, the stability of the inventory level greatly reduces the inaccuracy of predictions by the operator and improves the service quality accordingly.

Considering the stochastic characteristics of the NSICSA, GASA and SICSA, statistical tests ought to be performed. The one-way ANOVA test is adopted to analyze computational results based on problems of three scales. It is obvious that the algorithm section is the single factor in the one-way ANOVA under study, and the null hypothesis is that there is no significant difference about the algorithm section in different instances. Statistical results are shown in [Table VI](#), where the confidence level is set to 95 per cent as normal. Consequently, the null hypothesis will be rejected if

Table VI One-way ANOVA table

Scale	Source	Degree of freedom	Sum of squares	Mean of squares	F-value	P-value
Small	Factor	2	257	128	0.01	0.993
	Error	57	1,010,643	17,731		
	Total	59	1,010,900			
Medium	Factor	2	21,676	10,838	0.08	0.925
	Error	57	7,883,815	138,313		
	Total	59	7,905,492			
Large	Factor	2	4,872,445	2,436,222	3.18	0.049
	Error	57	43,682,125	766,353		
	Total	59	48,554,570			

Figure 10 Ladder diagram of inventory level of the first workstation



the P -value is less than 0.05 and the F -value is much larger than 1, that is to say, the different algorithm has a significant impact on the observed solutions from a statistical perspective. It can be seen from the table that the proposed NSICSA exceeds the GASA and SICSA in a statistically significant manner in the large-scale instance, which is usually the real-world scale instance because the P -value (0.049) meets the rejection condition.

5.3 Inventory level of each work cycles

Figure 10 shows that the ladder diagram of the inventory level of the first workstation under the moderate problem scale. It can be seen from the figure that because of the capacity limitation of the inventory of workstations, the inventory level of the station in each cycle is kept within 3, which is a relatively low level. For the two adjacent delivery jobs of the same workstation, which are denoted by φ_1 , φ_2 , the local minimum of inventory level in the workstation s is reached at the time of $t = t_{\varphi_2} + p_s - 1$, and the local maximum of inventory level in the workstation s is reached at the time of $t = t_{\varphi_1} + p_s$ as described previously in Theorem 2.

To prove the superiority of the inventory level obtained by the NSCISA algorithm, the improved GA (Xiong *et al.*, 2018) (IGA) is selected as the exact comparison. Table VII shows the inventory level of the first workstation in each cycle under the NSCISA algorithm and the IGA algorithm, as well as the gap between them.

According to the statistical analysis results in Table VII, among the 50 cases, there are 42 cases demonstrating that the inventory level obtained by the NSCISA is lower than that of the IGA. In the meanwhile, the upper limit of the inventory level under the NSCISA is 3 while the upper limit of the inventory level obtained by the IGA is 5. Therefore, it can be concluded that the NSCISA algorithm plays an effective and significant role in the inventory level reduction compared with the IGA.

5.4 Computational complexity

To evaluate the computational complexity, the average total runtime of each algorithm for different problem scales of all case tests are measured. The results are presented in Table VIII, which shows that owing to the advanced global

Table VII Inventory level under the NSCISA and the IGA

Work cycle	NSICSA	IGA	Gap	Whether superior
1	1	0	-1	N
2	3	4	1	Y
3	3	5	2	Y
4	1	3	2	Y
5	1	1	0	N
6	0	1	1	Y
7	0	0	0	N
8	3	3	0	N
9	2	3	1	Y
10	1	2	1	Y
11	0	2	2	Y
12	0	0	0	N
13	2	4	2	Y
14	2	3	1	Y
15	2	3	1	Y
16	1	3	2	Y
17	0	1	1	Y
18	0	1	1	Y
19	0	0	0	N
20	3	5	2	Y
21	3	5	2	Y
22	2	4	2	Y
23	2	1	-1	N
24	1	3	2	Y
25	1	2	1	Y
26	0	2	2	Y
27	0	1	1	Y
28	2	3	1	Y
29	2	1	-1	N
30	1	2	1	Y
31	0	1	1	Y
32	0	1	1	Y
33	0	5	5	Y
34	3	5	2	Y
35	3	2	-1	N
36	2	3	1	Y
37	2	3	1	Y
38	1	3	2	Y
39	1	2	1	Y
40	0	2	2	Y
41	2	5	3	Y
42	2	4	2	Y
43	2	4	2	Y
44	1	3	1	Y
45	1	2	1	Y
46	0	2	2	Y
47	3	5	2	Y
48	3	4	1	Y
49	2	4	2	Y
50	1	4	3	Y

exploitation ability, the proposed NSICSA does involve an additional computational cost that slightly affects the runtime. However, such an increase in runtime is slight and acceptable. In particular, the additional computational cost does actually decrease the average total inventory level and improve the

Table VIII Computational time of 60 instances

Group	(S , C)	$ S \cdot C$	NSICSA	GASA	SICSA
1	(5,9)	45	12.4	11.7	9.5
2	(5,18)	90	25.6	22.3	10.9
3	(5,27)	135	36.3	33.9	27.2
4	(5,36)	180	39.6	42.5	29.3
5	(10,9)	90	24.9	20.8	17.3
6	(10,18)	180	58.7	41.5	30.2
7	(10,27)	270	76.4	71.5	56.3
8	(10,36)	360	102.4	95.7	66.8
9	(15,9)	135	39.5	32.8	25.7
10	(15,18)	270	65.7	66.1	59.4
11	(15,27)	405	118.2	93.6	87.3
12	(15,36)	540	160.8	148.2	101.1
13	(20,9)	180	81.7	51.3	39.8
14	(20,18)	360	99.3	92.6	64.2
15	(20,27)	540	156.8	126.2	113.2
16	(20,36)	720	207.5	187.9	127.6
17	(25,9)	225	55.3	51.2	48.4
18	(25,18)	450	128.8	103.4	93.7
19	(25,27)	675	197.2	175.9	146.3
20	(25,36)	900	256.9	236.6	158.3
21	(30,45)	1,350	376.9	337.1	294.2
22	(30,54)	1,620	446.2	417.4	326.5
23	(30,63)	1,890	497.5	482.5	394.6
24	(30,72)	2,160	569.3	529.8	419.2
25	(35,45)	1,575	433.1	362.3	338.4
26	(35,54)	1,890	471.8	422.2	358.3
27	(35,63)	2,205	553.5	556.9	478.1
28	(35,72)	2,520	688.9	627.6	483.7
29	(40,45)	1,800	479.7	427.4	352.6
30	(40,54)	2,160	526.4	485.2	416.1
31	(40,63)	2,520	672.2	638.6	463.9
32	(40,72)	2,880	754.0	741.2	527.3
33	(45,45)	2,025	558.3	501.7	405.2
34	(45,54)	2,430	660.8	596.4	466.0
35	(45,63)	2,835	832.9	709.1	543.1
36	(45,72)	3,240	883.4	815.8	614.8
37	(50,45)	2,250	587.5	539.6	463.7
38	(50,54)	2,700	725.2	568.1	505.2
39	(50,63)	3,150	943.1	791.2	624.8
40	(50,72)	3,600	1,086.3	891.4	709.1
41	(55,81)	4,455	1,208.4	1,203.7	903.4
42	(55,90)	4,950	1,337.3	1,211.9	968.2
43	(55,99)	5,445	1,454.9	1,365.8	1,064.2
44	(55,108)	5,940	1,594.5	1,339.2	1,124.9
45	(60,81)	4,860	1,125.7	1,056.2	964.7
46	(60,90)	5,400	1,430.9	1,138.3	1,032.8
47	(60,99)	5,940	1,473.2	1,369.5	1,188.4
48	(60,108)	6,480	1,733.6	1,599.8	1,357.5
49	(65,81)	5,265	1,261.7	1,163.2	1,035.7
50	(65,90)	5,850	1,659.2	1,224.5	1,153.1
51	(65,99)	6,435	1,789.5	1,524.8	1,307.2
52	(65,108)	7,020	1,893.1	1,626.3	1,226.5
53	(70,81)	5,670	1,317.2	1,254.7	1,178.4
54	(70,90)	6,300	1,675.8	1,353.1	1,224.2
55	(70,99)	6,930	1,824.7	1,525.6	1,354.6

(continued)

Table VIII

Group	(S , C)	$ S \cdot C$	NSICSA	GASA	SICSA
56	(70,108)	7,560	2,019.4	1,669.8	1,539.7
57	(75,81)	6,075	1,513.2	1,362.3	1,213.2
58	(75,90)	6,750	1,796.9	1,668.3	1,275.9
59	(75,99)	7,425	1,927.1	1,816.9	1,499.3
60	(75,108)	8,100	2,088.5	2,037.1	1,809.8

quality of results according to the comparison between the proposed algorithm and the others in most realistic problem instances as is shown in Tables III–V. Consequently, it can be concluded that the computational complexity of NSICSA is worthwhile owing to its convergence performance and search ability.

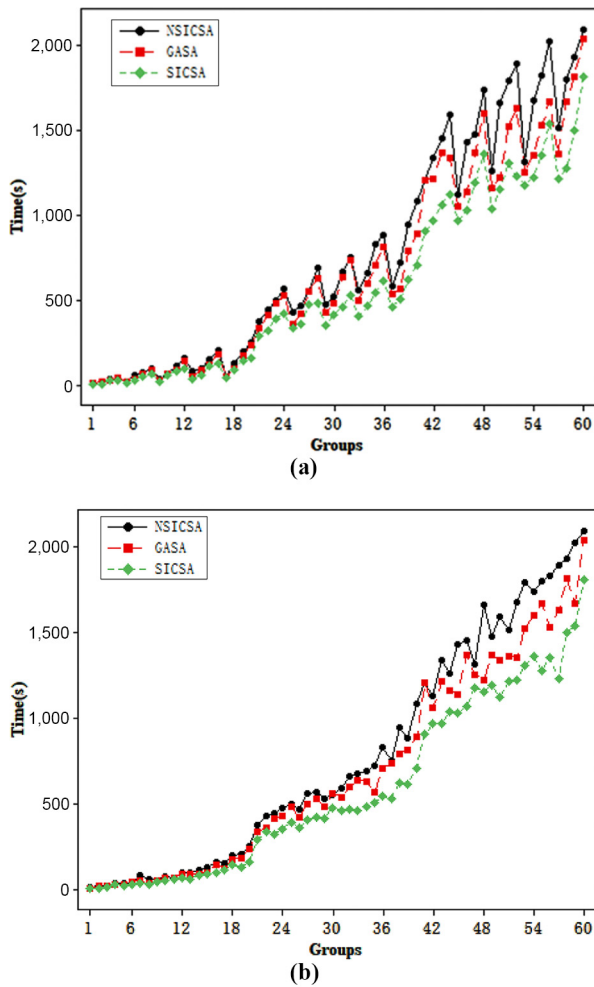
In addition, to better visualize the fluctuation pattern of computational time of three algorithms, the runtime can be observed in Figure 11(a), and the time arranged according to the values of $|S| \cdot C$ in an ascending order is shown in Figure 11(b). It can be confirmed that the computational time rises rapidly when problem scales and parameters increase significantly according to Table II, especially at the junction of medium-scale instances and large-scale instances. Larger scale problems show more computationally expensive performances owing to the larger number of iterations and antibodies, which provides the NSICSA with a greater diversity of solutions and leads to superior results. Hence, it can be drawn that the computational complexity of the NSICSA is acceptable.

5.5 Managerial applications

When the number of workstations is considerably large and the inventory capacity of each workstation needs to maintain a low level, given the limitation of loading capacity of tow trains, the speed of driving, loading and unloading and the time of material replenishment in the supermarket, a fairly comprehensive scheduling method may not meet the requirement. Consequently, it is necessary to increase the number of tow trains to carry out delivery tasks. The workstations need to be partitioned to reduce the driving time and transmit time of tow trains, so as to achieve an efficient delivery scheme. The cost of material inventory can be substantially reduced despite a slightly increase in the fixed cost. So, in the long run, the total cost can be saved. Accordingly, when implementing the comprehensive scheduling scheme, it is inevitable to consider the actual demand of the workshop and think over if the number of tow trains need to be increased, which will reduce the overall cost and obtain an efficient operation of the MMAL. Figure 12 plots the experimental results of total inventory when increasing the number of the tow train under a moderate problem scale.

This section provides real-situation scheduling operators and managers with feasible scheduling applications. When the inventory needs to maintain a certain low level and the number of workstations is large, the similar analysis can be used to determine the number of tow trains and inventory capacity limit so that the

Figure 11 Computational time of NSICSA, GASA and SICSA



Notes: (a); (b)

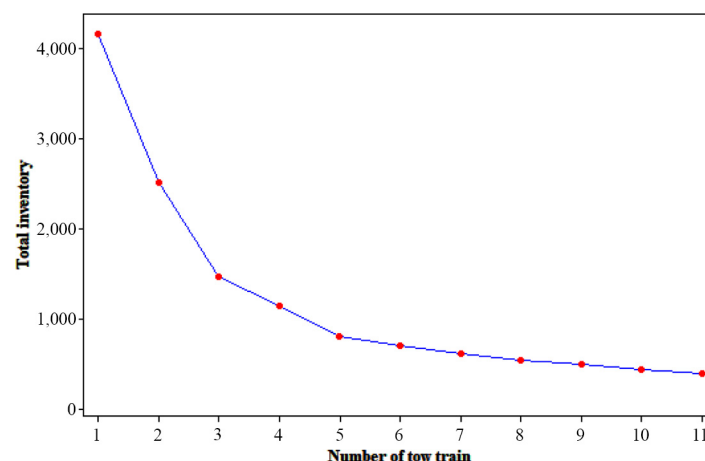
assembly line can gain the best production efficiency and economic benefits.

6. Conclusions

In this paper, we have investigated the scheduling and loading of tow trains of in-plant milk-run delivery for MMALs. A combined scheduling and loading distribution model was formulated to minimize total inventory for all workstations over the planning horizon. We proposed a novel optimal method NSICSA, consisting of the mechanism of ICSCA with the mechanism of SA and NS. The solution obtained provided us a suitable timetable for tow trains and the quantity of bins to be delivered to each workstation under the precedence constrain. It is observed that the NSICSA outperforms other methods on the performance of convergence and the quality of solutions, especially when the problem scale increases to the real-world scale. It is able to obtain solutions with lower variabilities within an acceptable computational time. Furthermore, it verifies the validity of the capacity limit of line-side inventory. The current work is capable of solving the scheduling and loading problems of in-plant milk-run delivery, and it can be modified to solve similar part feeding problems.

The problem discussed in this paper is based on the assumption that tow trains will not break down during the distribution process, and demands over the planning horizon is known, but uncertainties are likely to occur in the actual situation. Therefore, uncertain factors such as tow train breakdowns, short-term adjustments of production planning and downtime of the assembly line may be taken into account, that is to say, the future work will focus on the real-time scheduling which is dynamically scheduled according to the system status (such as the demand of each workstation, the tow train status, etc.), thus having the ability to cope well with the dynamic environment and having wide applications in the practical production.

Figure 12 Total inventory when increasing the number of tow train



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